

PolyLog Integrals

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1 PolyLog

PolyLog may be defined as the series below

$$\text{Li}_s(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^s}, |z| < 1$$

Or integral

$$\text{Li}_s(z) = \frac{z}{\Gamma(s)} \int_0^{\infty} \frac{x^{s-1}}{e^x - z} dx$$

1.1 Prerequisites

A

Let's review the well-known Gamma Function

$$\Gamma(z) = \int_0^{\infty} x^{z-1} e^{-x} dx, \Re(z) > 0$$

Integrate by parts we get:

$$\Gamma(z+1) = z\Gamma(z)$$

From the Dominated convergence theorem, Gamma Function could be denoted as

$$\Gamma(z) = \lim_{n \rightarrow \infty} \int_0^n \left(1 - \frac{x}{n}\right)^n x^{z-1} dx$$

Simplify after recurring IBP, we have Weierstrass's product form:

$$\Gamma(z) = \frac{1}{z} e^{-\gamma z} \prod_{n=1}^{\infty} \left(\left(1 + \frac{z}{n}\right)^{-1} e^{\frac{z}{n}} \right)$$

Euler's Reflection Formula:

$$\Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}$$

It's not hard to prove Legendre's Duplication Formula by the trigonometric form of Beta Function:

$$\Gamma(2z) = \frac{2^{2z-1}}{\sqrt{\pi}} \Gamma(z) \Gamma\left(z + \frac{1}{2}\right)$$

Beta Function:

$$B(p, q) = \int_0^1 x^{p-1} (1-x)^{q-1} dx = 2 \int_0^{\frac{\pi}{2}} \sin^{2p-1} x \cos^{2q-1} x dx, \Re(p) > 0, \Re(q) > 0$$

Which is closely associated with Gamma Function:

$$B(p, q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$$

PolyGamma is defined by the nth derivative of the logarithm of Gamma Function:

$$\psi(z) = \frac{d}{dz} \log \Gamma(z) = \frac{\Gamma'(z)}{\Gamma(z)}$$

Series representation:

$$\begin{aligned}\psi(z) &= -\gamma - \frac{1}{z} + \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+z} \right) \\ \psi^{(n)}(z) &= (-1)^{n+1} n! \sum_{k=0}^{\infty} \frac{1}{(z+k)^{n+1}}\end{aligned}$$

Integral representation:

$$\psi(z) = -\gamma + \int_0^1 \frac{1-x^{z-1}}{1-x} dx$$

B

Series

$$\begin{aligned}\zeta(z) &= \sum_{n=1}^{\infty} \frac{1}{n^z}, \Re(z) > 1 \\ \eta(z) &= \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^z}, \Re(z) > 1 \\ \lambda(z) &= \sum_{n=1}^{\infty} \frac{1}{(2n-1)^z}, \Re(z) > 0 \\ \beta(z) &= \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(2n-1)^z}, \Re(z) > 0\end{aligned}$$

They are commonly applied to integrals

$$\begin{aligned}\int_0^{\infty} \frac{x^{z-1}}{e^x - 1} dx &= \Gamma(z) \zeta(z) \\ \int_0^{\infty} \frac{x^{z-1}}{e^x + 1} dx &= \Gamma(z) \eta(z) \\ \int_0^{\infty} \frac{x^{z-1}}{e^x - e^{-x}} dx &= \Gamma(z) \lambda(z) \\ \int_0^{\infty} \frac{x^{z-1}}{e^x + e^{-x}} dx &= \Gamma(z) \beta(z)\end{aligned}$$

As well as Bernoulli and Euler numbers

$$\begin{aligned}\zeta(2n) &= \frac{(-1)^{n-1} (2\pi)^{2n}}{2(2n)!} B_{2n} \\ \beta(2n+1) &= \frac{(-1)^n \pi^{2n+1}}{4^{n+1} (2n)!} E_{2n}\end{aligned}$$

C

Fourier Series

$$\begin{aligned}\log \sin x &= -\log 2 - \sum_{n=1}^{\infty} \frac{\cos 2nx}{n} \\ \log \cos x &= -\log 2 - \sum_{n=1}^{\infty} \frac{(-1)^n \cos 2nx}{n}\end{aligned}$$

The goal is to compute the Fourier series of $g(x) = \log \sin x$ over $[0, \pi]$, or the Fourier series of $h(x) = \log \sin \frac{x}{2}$ over $[0, 2\pi]$, or the Fourier series of $f(x) = \log \cos \frac{x}{2}$ over $[-\pi, \pi]$. Since $f(x)$ is an even function, we have to compute:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{+\pi} \cos(nx) \log \cos \frac{x}{2} dx = \frac{2}{\pi} \int_0^{\pi} \cos(nx) \log \cos \frac{x}{2} dx$$

for any $n \geq 1$ to be able to state:

$$f(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx + \sum_{n \geq 1} a_n \cos(nx) = \frac{a_0}{2} + \sum_{n \geq 1} a_n \cos(nx).$$

for any $x \in (-\pi, \pi)$. Integration by parts gives:

$$a_n = \frac{2}{\pi} \left(\frac{1}{n} \sin(nx) \log \cos \frac{x}{2} \Big|_0^{\pi} + \frac{1}{2n} \int_0^{\pi} \sin(nx) \tan \frac{x}{2} dx \right)$$

or just:

$$a_n = \frac{1}{\pi n} \int_0^{\pi} \frac{\sin(nx) \sin(x/2)}{\cos(x/2)} dx = -\frac{2}{\pi n} \int_0^{\frac{\pi}{2}} \frac{\cos((2n+1)x)}{\cos x} dx.$$

Since $\cos((2n+1)x) = 2 \cos x \cos(2nx) - \cos((2n-1)x)$, we have:

$$\int_0^{\frac{\pi}{2}} \frac{\cos((2n+1)x)}{\cos x} dx = - \int_0^{\frac{\pi}{2}} \frac{\cos((2n-1)x)}{\cos x} dx = (-1)^n \frac{\pi}{2}.$$

This gives:

$$\log \cos \frac{x}{2} = \frac{a_0}{2} + \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \cos(nx)$$

for any $x \in (-\pi, \pi)$. In order to find a_0 , we can simply match $f(0) = 0$ with the series on the right hand side. Since:

$$\sum_{n \geq 1} \frac{(-1)^{n+1}}{n} = \int_0^1 \frac{dx}{1+x} = \log 2,$$

we have:

$$\log \cos \frac{x}{2} = -\log 2 + \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \cos(nx) \quad \forall x \in (-\pi, \pi)$$

and by translating the variable:

$$\log \sin \frac{x}{2} = -\log 2 - \sum_{n \geq 1} \frac{1}{n} \cos(nx) \quad \forall x \in (0, 2\pi)$$

$$\log \sin x = -\log 2 - \sum_{n \geq 1} \frac{1}{n} \cos(2nx) \quad \forall x \in (0, \pi)$$

as wanted.

In fact proving from the right hand side is much clearer

$$\begin{aligned} -\log 2 - \sum_{n=1}^{\infty} \frac{\cos 2nx}{n} &= -\log 2 - \Im \sum_{n=1}^{\infty} \frac{e^{2inx}}{n} \\ &= -\log 2 - \Im \text{Li}_1(e^{2ix}) \\ &= -\log 2 + \Im \log(1 - e^{2ix}) \\ &= \log \sin x \end{aligned}$$

From the series we have

$$\int_0^{\frac{\pi}{2}} \log \sin x dx = \int_0^{\frac{\pi}{2}} \log \cos x dx = -\frac{\pi}{2} \log 2$$

Similarly we have the Catalan Constant

$$G = \int_0^{\frac{\pi}{4}} \log \cot x dx = \beta(2) = 0.915965594 \dots$$

1.2 Properties

A

It's important to be acquainted with these specific formulas

$$\begin{aligned} \text{Li}_0(x) &= \frac{x}{1-x} \\ \text{Li}_1(x) &= -\log(1-x) \\ \text{Li}_2(x) + \text{Li}_2\left(\frac{x}{x-1}\right) &= -\frac{1}{2} \log^2(1-x) \\ \text{Li}_2(x) + \text{Li}_2(-x) &= \frac{1}{2} \text{Li}_2(x^2) \\ \text{Li}_2(x) + \text{Li}_2(1-x) &= \frac{\pi^2}{6} - \log x \log(1-x) \end{aligned}$$

Take the third for example. To prove it, we simply need to differentiate on both sides:

$$\begin{aligned} -\frac{\log(1-x)}{x} + \frac{\log(1-x)}{x} + \frac{\log(1-x)}{1-x} &= \frac{\log(1-x)}{1-x} \\ \text{Li}_2(x) + \text{Li}_2\left(\frac{x}{x-1}\right) &= -\frac{1}{2} \log^2(1-x) + C \end{aligned}$$

Plug $x = 0$ we get the demanded form. The other identities can be proved using the same method.

B

$$\begin{aligned}\mathrm{Li}_n(x) &= \int_0^x \frac{\mathrm{Li}_{n-1}(x)}{x} dx \\ \mathrm{Li}_n(x) &= \frac{(-1)^{n-1}}{(n-1)!} \int_0^1 \frac{x \log^{n-1} y}{1-xy} dy\end{aligned}$$

Let's prove them quickly:

$$\begin{aligned}\int_0^x \frac{\mathrm{Li}_{n-1}(x)}{x} dx &= \int_0^x \frac{\sum_{k=1}^{\infty} \frac{x^k}{k^{n-1}}}{x} dx \\ &= \sum_{k=1}^{\infty} \frac{1}{k^{n-1}} \int_0^x x^{k-1} dx \\ &= \sum_{k=1}^{\infty} \frac{x^k}{k^n} \\ &= \mathrm{Li}_n(x) \\ \frac{(-1)^{n-1}}{(n-1)!} \int_0^1 \frac{x \log^{n-1} y}{1-xy} dy &= \frac{(-1)^{n-1}}{(n-1)!} \int_0^1 x \log^{n-1} y \sum_{k=0}^{\infty} (xy)^k dy \\ &= \frac{(-1)^{n-1}}{(n-1)!} \sum_{k=0}^{\infty} x^{k+1} \underbrace{\int_0^1 y^k \log^{n-1} y dy}_{y \rightarrow e^{-\frac{y}{k+1}}} \\ &= \frac{1}{(n-1)!} \sum_{k=0}^{\infty} \frac{x^{k+1}}{(k+1)^n} \int_0^{\infty} e^{-y} y^{n-1} dy \\ &= \frac{1}{(n-1)!} \sum_{k=1}^{\infty} \frac{x^k}{k^n} \Gamma(n) \\ &= \mathrm{Li}_n(x)\end{aligned}$$

1.3 Practices

A001

$$\mathrm{Li}_2(i) = -\frac{\pi^2}{48} + iG$$

Proof

$$\begin{aligned}\mathrm{Li}_2(i) &= \sum_{n=1}^{\infty} \frac{i^n}{n^2} \\ &= \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)^2} + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}i}{(2n-1)^2} \\ &= -\frac{\pi^2}{48} + iG\end{aligned}$$

A002

$$\text{Li}_2\left(\frac{1}{1+i}\right) = \frac{5\pi^2}{96} - \frac{\log^2 2}{8} + i\left(\frac{\pi}{8} \log 2 - G\right)$$

Proof

Apply

$$\text{Li}_2(x) + \text{Li}_2\left(\frac{x}{x-1}\right) = -\frac{1}{2} \log^2(1-x)$$

We have

$$\begin{aligned} \text{Li}_2\left(\frac{1-i}{2}\right) + \text{Li}_2(i) &= -\frac{1}{2} \log^2\left(\frac{1+i}{2}\right) \\ \text{Li}_2\left(\frac{1-i}{2}\right) - \frac{\pi^2}{48} + iG &= -\frac{1}{8} \log^2 2 + \frac{\pi^3}{32} + i\frac{\pi \log 2}{8} \\ \text{Li}_2\left(\frac{1}{1+i}\right) &= \frac{5\pi^2}{96} - \frac{\log^2 2}{8} + i\left(\frac{\pi}{8} \log 2 - G\right) \end{aligned}$$

A003

$$\text{Li}_2\left(\frac{\sqrt{5}-1}{2}\right) = \frac{\pi^2}{10} - \log^2\left(\frac{1+\sqrt{5}}{2}\right)$$

Proof

Apply

$$\begin{aligned} \text{Li}_2(x) + \text{Li}_2\left(\frac{x}{x-1}\right) &= -\frac{1}{2} \log^2(1-x) \\ \text{Li}_2(x) + \text{Li}_2(-x) &= \frac{1}{2} \text{Li}_2(x^2) \\ \text{Li}_2(x) + \text{Li}_2(1-x) &= \frac{\pi^2}{6} - \log x \log(1-x) \end{aligned}$$

We have

$$\begin{aligned} \text{Li}_2\left(\frac{3-\sqrt{5}}{2}\right) + \text{Li}_2\left(\frac{1-\sqrt{5}}{2}\right) &= -\frac{1}{2} \log^2\left(\frac{1+\sqrt{5}}{2}\right) \\ \text{Li}_2\left(\frac{\sqrt{5}-1}{2}\right) + \text{Li}_2\left(\frac{1-\sqrt{5}}{2}\right) &= \frac{1}{2} \text{Li}_2\left(\frac{3-\sqrt{5}}{2}\right) \\ \text{Li}_2\left(\frac{\sqrt{5}-1}{2}\right) + \text{Li}_2\left(\frac{3-\sqrt{5}}{2}\right) &= \frac{\pi^2}{6} - 2 \log^2\left(\frac{1+\sqrt{5}}{2}\right) \end{aligned}$$

Solve this system of equations

$$\text{Li}_2\left(\frac{\sqrt{5}-1}{2}\right) = \frac{\pi^2}{10} - \log^2\left(\frac{1+\sqrt{5}}{2}\right)$$

A004

$$\int_0^1 \frac{\log(1+x^2)}{x} dx = \frac{\pi^2}{12}$$

Proof

$$\begin{aligned}\int_0^1 \frac{\log(1+x^2)}{x} dx &\stackrel{x^2 \rightarrow x}{=} \frac{1}{2} \int_0^1 \frac{\log(1+x)}{x} dx \\ &= -\frac{1}{2} \text{Li}_2(-1) \\ &= \frac{\pi^2}{12}\end{aligned}$$

A005

$$\int_0^1 \frac{\log^2(1+x^2)}{x^3} dx = \frac{\pi^2}{12} - \log^2 2$$

Proof

$$\begin{aligned}\int_0^1 \frac{\log^2(1+x^2)}{x^3} dx &= \int_0^1 \log^2(1+x^2) d\left(-\frac{1}{2x^2}\right) \\ &= -\frac{1}{2} \log^2 2 + 2 \int_0^1 \left(\frac{\log(1+x^2)}{x} - \frac{x \log(1+x^2)}{1+x^2}\right) dx \\ &= \frac{\pi^2}{12} - \log^2 2\end{aligned}$$

A006

$$\begin{aligned}\int_0^{\frac{\pi}{4}} \log \cos x dx &= \frac{G}{2} - \frac{\pi}{4} \log 2 \\ \int_0^{\frac{\pi}{4}} \log \sin x dx &= -\frac{G}{2} - \frac{\pi}{4} \log 2\end{aligned}$$

Proof

Set

$$\begin{aligned}I_1 &= \int_0^{\frac{\pi}{4}} \log \cos x dx \\ I_2 &= \int_0^{\frac{\pi}{4}} \log \sin x dx\end{aligned}$$

Then

$$\begin{aligned}I_1 + I_2 &= \int_0^{\frac{\pi}{4}} \log\left(\frac{1}{2} \sin 2x\right) dx \\ &= -\frac{\pi}{4} \log 2 + \frac{1}{2} \int_0^{\frac{\pi}{2}} \log \sin x dx \\ &= -\frac{\pi}{2} \log 2 \\ I_1 - I_2 &= \int_0^{\frac{\pi}{4}} \log \cot x dx \\ &= G\end{aligned}$$

Solve this system of equations

$$\begin{aligned}\int_0^{\frac{\pi}{4}} \log \cos x dx &= \frac{G}{2} - \frac{\pi}{4} \log 2 \\ \int_0^{\frac{\pi}{4}} \log \sin x dx &= -\frac{G}{2} - \frac{\pi}{4} \log 2\end{aligned}$$

A007

$$\int_0^{\frac{\pi}{4}} x \log \cos x dx = \frac{\pi G}{8} - \frac{\pi^2}{32} \log 2 - \frac{21}{128} \zeta(3)$$

Proof

$$\begin{aligned}\int_0^{\frac{\pi}{4}} x \log \cos x dx &= - \int_0^{\frac{\pi}{4}} x \left(\log 2 + \sum_{n=1}^{\infty} \frac{(-1)^n \cos 2nx}{n} \right) dx \\ &= - \log 2 \int_0^{\frac{\pi}{4}} x dx - \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \int_0^{\frac{\pi}{4}} x \cos 2nx dx \\ &= - \frac{\pi^2}{32} \log 2 - \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \left(\frac{\pi \sin(\frac{n\pi}{2})}{8n} + \frac{\cos(\frac{n\pi}{2})}{4n^2} - \frac{1}{4n^2} \right) \\ &= - \frac{\pi^2}{32} \log 2 - \frac{\pi}{8} \sum_{n=1}^{\infty} \frac{(-1)^n \sin(\frac{n\pi}{2})}{n^2} - \frac{1}{4} \sum_{n=1}^{\infty} \frac{(-1)^n \cos(\frac{n\pi}{2})}{n^3} + \frac{1}{4} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^3} \\ &= - \frac{\pi^2}{32} \log 2 - \frac{\pi}{8} \Im \text{Li}_2(-i) - \frac{1}{4} \Re \text{Li}_3(-i) + \frac{1}{4} \text{Li}_3(-1) \\ &= \frac{\pi G}{8} - \frac{\pi^2}{32} \log 2 - \frac{21}{128} \zeta(3)\end{aligned}$$

A008

$$\int_0^{\frac{\pi}{4}} x \log(1 + \tan x) dx = \frac{\pi^2}{64} \log 2 - \frac{\pi G}{8} + \frac{21\zeta(3)}{64}$$

Proof

$$\begin{aligned}
\int_0^{\frac{\pi}{4}} x \log(1 + \tan x) dx &= \int_0^{\frac{\pi}{4}} x \log(\sqrt{2} \sin(x + \frac{\pi}{4})) dx - \int_0^{\frac{\pi}{4}} x \log \cos x dx \\
&= \frac{\pi^2}{64} \log 2 + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (x - \frac{\pi}{4}) \log \sin x dx - \int_0^{\frac{\pi}{4}} x \log \cos x dx \\
&= \frac{\pi^2}{64} \log 2 + \int_0^{\frac{\pi}{4}} (\frac{\pi}{4} - x) \log \cos x dx - \int_0^{\frac{\pi}{4}} x \log \cos x dx \\
&= \frac{\pi^2}{64} \log 2 + \frac{\pi}{4} \int_0^{\frac{\pi}{4}} \log \cos x dx - 2 \int_0^{\frac{\pi}{4}} x \log \cos x dx \\
&= \frac{\pi^2}{64} \log 2 - \frac{\pi G}{8} + \frac{21\zeta(3)}{64}
\end{aligned}$$

A009

$$\int_0^{\frac{\pi}{2}} x^2 \cot x dx = \frac{1}{4} \pi^2 \log 2 - \frac{7}{8} \zeta(3)$$

Proof

Consider

$$\cot x = -\frac{i(e^{-ix} + e^{ix})}{e^{-ix} - e^{ix}} = -i - 2i\text{Li}_0(e^{2ix})$$

We have

$$\begin{aligned}
\int x^2 \cot x dx &= -i \int x^2 dx - 2i \int x^2 \text{Li}_0(e^{2ix}) dx \\
&= -\frac{ix^3}{3} - \int x^2 d(\text{Li}_1(e^{2ix})) \\
&= -\frac{ix^3}{3} - x^2 \text{Li}_1(e^{2ix}) + 2 \int x \text{Li}_1(e^{2ix}) dx \\
&= -\frac{ix^3}{3} - x^2 \text{Li}_1(e^{2ix}) - ix \text{Li}_2(e^{2ix}) + i \int \text{Li}_2(e^{2ix}) dx \\
&= -\frac{ix^3}{3} - x^2 \text{Li}_1(e^{2ix}) - ix \text{Li}_2(e^{2ix}) + \frac{1}{2} \text{Li}_3(e^{2ix}) + C
\end{aligned}$$

Hence

$$\int_0^{\frac{\pi}{2}} x^2 \cot x dx = \frac{1}{4} \pi^2 \log 2 - \frac{7}{8} \zeta(3)$$

A010

$$\int_0^{\frac{\pi}{4}} x^2 \tan^2 x dx = -G + \frac{\pi^2}{16} - \frac{\pi^3}{192} + \frac{\pi}{4} \log 2$$

Proof

Consider

$$\begin{aligned}
(x^2 \tan x)' &= 2x \tan x + x^2 \sec^2 x \\
&= x^2 + 2x \tan x + x^2 \tan^2 x \\
\int x^2 \tan^2 x dx &= x^2 \tan x - \int x^2 dx - 2 \int x \tan x dx
\end{aligned}$$

We have

$$\begin{aligned}
\int_0^{\frac{\pi}{4}} x^2 \tan^2 x dx &= x^2 \tan x - \frac{x^3}{3} - 2i \int x(1 + 2\text{Li}_0(-e^{2ix})) dx \\
&= x^2 \tan x - \frac{x^3}{3} - ix^2 - 2 \int x d(\text{Li}_1(-e^{2ix})) \\
&= x^2 \tan x - \frac{x^3}{3} - ix^2 - 2x\text{Li}_1(-e^{2ix}) - i\text{Li}_2(-e^{2ix}) + C
\end{aligned}$$

Hence

$$\int_0^{\frac{\pi}{4}} x^2 \tan^2 x dx = -G + \frac{\pi^2}{16} - \frac{\pi^3}{192} + \frac{\pi}{4} \log 2$$

A011

$$\int_0^1 \frac{\text{Li}_2(x)}{x\sqrt{1-x^2}} dx = \frac{3}{8}\pi^2 \log 2 - \frac{7}{16}\zeta(3)$$

Proof

$$\begin{aligned}
\int_0^1 \frac{\text{Li}_2(x)}{x\sqrt{1-x^2}} dx &= - \int_0^1 \int_0^1 \frac{1}{x\sqrt{1-x^2}} \frac{x \log y}{1-xy} dy dx \\
&= - \int_0^1 \log y \int_0^1 \frac{1}{(1-xy)\sqrt{1-x^2}} dx dy \\
&= - \int_0^1 \frac{\pi - \arccos y}{\sqrt{1-y^2}} \log y dy \\
&= - \int_0^1 \frac{\pi - \frac{\pi}{2} + \arcsin x}{\sqrt{1-y^2}} \log y dy \\
&= -\frac{\pi}{2} \int_0^1 \frac{\log y}{\sqrt{1-y^2}} dy - \int_0^1 \frac{\arcsin x}{\sqrt{1-y^2}} \log y dy \\
&= \frac{1}{4}\pi^2 \log 2 + \frac{1}{2} \int_0^1 \frac{\arcsin^2 y}{y} dy \\
&= \frac{1}{4}\pi^2 \log 2 + \frac{1}{2} \int_0^{\frac{\pi}{2}} y^2 \cot y dy \\
&= \frac{3}{8}\pi^2 \log 2 - \frac{7}{16}\zeta(3)
\end{aligned}$$

A012

$$\int_0^1 \frac{\arctan x}{x+1} \log x dx = -\frac{\pi^3}{64} + \frac{G}{2} \log 2$$

Proof

$$\begin{aligned} \text{LHS} &= \int_0^1 \arctan x d(\text{Li}_2(-x) + \log x \log(x+1)) \\ &= -\frac{\pi^3}{48} - \int_0^1 \frac{\log x \log(x+1)}{x^2+1} dx - \int_0^1 \frac{\text{Li}_2(-x)}{x^2+1} dx \\ &= -\frac{\pi^3}{48} - \int_0^1 \frac{\log x \log(x+1)}{x^2+1} dx - \int_0^1 \frac{1}{x^2+1} \int_0^1 \frac{x \log y}{xy+1} dy dx \\ &= -\frac{\pi^3}{48} - \int_0^1 \frac{\log x \log(x+1)}{x^2+1} dx - \int_0^1 \log y \int_0^1 \frac{x}{(x^2+1)(xy+1)} dx dy \\ &= -\frac{\pi^3}{48} - \int_0^1 \frac{\log x \log(x+1)}{x^2+1} dx - \int_0^1 \log y \left(\frac{\pi y}{4(y^2+1)} + \frac{\log 2}{2(y^2+1)} - \frac{\log(y+1)}{y^2+1} \right) dy \\ &= -\frac{\pi^3}{48} - \underbrace{\frac{\pi}{4} \int_0^1 \frac{y \log y}{y^2+1} dy}_{y \rightarrow e^{-y}} - \frac{\log 2}{2} \underbrace{\int_0^1 \frac{\log y}{y^2+1} dy}_{y \rightarrow e^{-y}} \\ &= -\frac{\pi^3}{48} + \frac{\pi}{4} \int_0^\infty \frac{y}{e^{2y}+1} dy + \frac{\log 2}{2} \int_0^\infty \frac{x}{e^x + e^{-x}} dx \\ &= -\frac{\pi^3}{48} + \frac{\pi}{16} \eta(2) \Gamma(2) + \frac{\log 2}{2} \beta(2) \Gamma(2) \\ &= -\frac{\pi^3}{64} + \frac{G}{2} \log 2 \\ &= \text{RHS} \end{aligned}$$

A013

$$\int_0^1 \frac{\arctan x}{x+1} \log^2 x dx = -\frac{\pi^2}{24} G - \frac{\pi^3}{32} \log 2 + \frac{21\pi}{64} \zeta(3)$$

Proof

$$\begin{aligned} \text{LHS} &= \int_0^1 \arctan x d(-2\text{Li}_3(-x) + 2\log x \text{Li}_2(-x) + \log(x+1) \log^2 x) \\ &= \frac{3\pi}{8} \zeta(3) + \int_0^1 \frac{2\text{Li}_3(-x) - 2\log x \text{Li}_2(-x) - \log(x+1) \log^2 x}{x^2+1} dx \\ &= \frac{3\pi}{8} \zeta(3) - \underbrace{\int_0^1 \int_0^1 \frac{x \log^2 y}{(xy+1)(x^2+1)} dy dx}_{\mathcal{J}} - 2 \underbrace{\int_0^1 \int_0^1 \frac{x \log x \log y}{(xy+1)(x^2+1)} dy dx}_{\mathcal{K}} - \underbrace{\int_0^1 \frac{\log(x+1) \log^2 x}{x^2+1} dx}_{\mathcal{M}} \end{aligned}$$

$$\begin{aligned}
\mathcal{J} &= \int_0^1 \log^2 y \left(\frac{\pi y}{4(y^2+1)} + \frac{\log 2}{2(y^2+1)} - \frac{\log(y+1)}{y^2+1} \right) dy \\
&= \frac{\pi}{4} \underbrace{\int_0^1 \frac{y \log^2 y}{y^2+1} dy}_{y \rightarrow e^{-y}} + \frac{\log 2}{2} \underbrace{\int_0^1 \frac{\log^2 y}{y^2+1} dy}_{y \rightarrow e^{-y}} - \int_0^1 \frac{\log^2 y \log(y+1)}{y^2+1} dy \\
&= \frac{\pi}{4} \int_0^\infty \frac{y^2}{e^{2y}+1} dy + \frac{\log 2}{2} \int_0^\infty \frac{y^2}{e^y + e^{-y}} dy - \mathcal{M} \\
&= \frac{\pi}{32} \eta(3) \Gamma(3) + \frac{\log 2}{2} \beta(3) \Gamma(3) - \mathcal{M} \\
&= \frac{3\pi}{64} \zeta(3) + \frac{\pi^3}{32} \log 2 - \mathcal{M} \\
\mathcal{K} &= \int_0^1 \log y \int_0^1 \log x \left(\frac{x+y}{(x^2+1)(y^2+1)} - \frac{y}{(xy+1)(y^2+1)} \right) dx dy \\
&= \int_0^1 \log y \left(\frac{1}{y^2+1} \int_0^1 \frac{x \log x}{x^2+1} dx + \frac{y}{y^2+1} \int_0^1 \frac{\log x}{x^2+1} dx - \frac{y}{y^2+1} \underbrace{\int_0^1 \frac{\log x}{xy+1} dx}_{\mathcal{N}} \right) dy \\
\mathcal{N} &= \int_0^1 \log x \sum_{n=0}^\infty (-1)^n (xy)^n dx \\
&= \sum_{n=0}^\infty (-y)^n \int_0^1 x^n \log x dx \\
&= - \sum_{n=0}^\infty \frac{(-y)^n}{(n+1)^2} \\
&= \frac{1}{y} \text{Li}_2(-y) \\
\mathcal{K} &= -\frac{\pi^2}{48} \int_0^1 \frac{\log y}{y^2+1} dy - G \int_0^1 \frac{y \log y}{y^2+1} dy - \int_0^1 \frac{\log y \text{Li}_2(-y)}{y^2+1} dy \\
&= \frac{\pi^2}{24} G - \int_0^1 \frac{\log y}{y^2+1} \int_0^1 \frac{y \log x}{xy+1} dx dy \\
&= \frac{\pi^2}{24} G - \mathcal{K} \\
\mathcal{K} &= \frac{\pi^2}{48} G \\
\text{LHS} &= \frac{3\pi}{8} \zeta(3) - \mathcal{J} - 2\mathcal{K} - \mathcal{M} \\
&= -\frac{\pi^2}{24} G - \frac{\pi^3}{32} \log 2 + \frac{21\pi}{64} \zeta(3) \\
&= \text{RHS}
\end{aligned}$$

A014

$$\int_0^1 \frac{\log x \log(x+1)}{x^2+1} dx = 2\Im(\text{Li}_3(1+i)) - \Im(\text{Li}_3(\frac{1+i}{2})) - 2G \log 2 - \frac{3\pi \log^2 2}{32} - \frac{3\pi^3}{128}$$

Proof

$$\begin{aligned} \text{LHS} &= -\frac{1}{2} \underbrace{\int_0^1 \frac{\log^2(\frac{x}{x+1})}{x^2+1} dx}_{\mathcal{J}} + \frac{1}{2} \underbrace{\int_0^1 \frac{\log^2 x}{x^2+1} dx}_{\mathcal{K}} + \frac{1}{2} \underbrace{\int_0^1 \frac{\log^2(x+1)}{x^2+1} dx}_{\mathcal{M}} \\ \mathcal{J} &\stackrel{\frac{x}{x+1} \rightarrow x}{=} \int_0^{\frac{1}{2}} \frac{\log^2 x}{2x^2-2x+1} dx \\ &= \Im \int_0^{\frac{1}{2}} \frac{\log^2 x}{x - \frac{1+i}{2}} dx \\ &= -\Im \int_0^{\frac{1}{2}} \frac{2 \log^2 x}{1+i} \sum_{n=0}^{\infty} \left(\frac{2x}{1+i}\right)^n dx \\ &= -\Im \sum_{n=0}^{\infty} \left(\frac{2}{1+i}\right)^{n+1} \int_0^{\frac{1}{2}} x^n \log^2 x dx \\ &= -\Im \sum_{n=0}^{\infty} \left(\frac{2}{1+i}\right)^{n+1} \left(\frac{x^{n+1} \log^2 x}{n+1} - \frac{2x^{n+1} \log x}{(n+1)^2} + \frac{2x^{n+1}}{(n+1)^3} \right) \Big|_0^{\frac{1}{2}} \\ &= -\Im \left(\log^2 2 \sum_{n=0}^{\infty} \frac{(\frac{1}{1+i})^{n+1}}{n+1} + 2 \log 2 \sum_{n=0}^{\infty} \frac{(\frac{1}{1+i})^{n+1}}{(n+1)^2} + 2 \sum_{n=0}^{\infty} \frac{(\frac{1}{1+i})^{n+1}}{(n+1)^3} \right) \\ &= -\Im \left(\log^2 2 \text{Li}_1\left(\frac{1}{1+i}\right) + 2 \log 2 \text{Li}_2\left(\frac{1}{1+i}\right) + 2 \text{Li}_3\left(\frac{1}{1+i}\right) \right) \\ &= 2G \log 2 + 2\Im(\text{Li}_3(\frac{1+i}{2})) \\ \mathcal{K} &= \Im \int_0^1 \frac{\log^2 x}{x-i} dx \\ &= -\Im \left(-i \int_0^1 \frac{\log^2 x}{1 - \frac{x}{i}} dx \right) \\ &= -\Im \left(\sum_{n=0}^{\infty} (-i)^{n+1} \int_0^1 x^n \log^2 x dx \right) \\ &= -\Im(2\text{Li}_3(-i)) \\ &= \frac{\pi^3}{16} \end{aligned}$$

$$\begin{aligned}
\mathcal{M} &= \Im \int_0^1 \frac{\log^2(1+x)}{x-i} dx \\
&\stackrel{x+1 \rightarrow x}{=} \Im \int_1^2 \frac{\log^2 x}{x-i} dx \\
&= -\Im \left(\frac{1}{1+i} \int_1^2 \frac{\log^2 x}{1-\frac{x}{1+i}} dx \right) \\
&= -\Im \left(\sum_{n=0}^{\infty} \left(\frac{1}{1+i} \right)^{n+1} \int_1^2 x^n \log^2 x dx \right) \\
&= 4\Im(\text{Li}_3(1+i)) - \frac{7\pi^3}{64} - \frac{3\pi \log^2 2}{16} - 2G \log 2 \\
\text{LHS} &= -\frac{1}{2}\mathcal{J} + \frac{1}{2}\mathcal{K} + \frac{1}{2}\mathcal{M} \\
&= 2\Im(\text{Li}_3(1+i)) - \Im(\text{Li}_3(\frac{1+i}{2})) - 2G \log 2 - \frac{3\pi \log^2 2}{32} - \frac{3\pi^3}{128} \\
&= \text{RHS}
\end{aligned}$$

A015

$$\int_0^{\frac{\pi}{2}} x \log^2 \sin x dx = \text{Li}_4\left(\frac{1}{2}\right) - \frac{19\pi^4}{2880} + \frac{1}{24} \log^4 2 + \frac{1}{12} \pi^2 \log^2 2$$

Proof

Introduce a dual integral for its simpler form:

$$\int_0^{\frac{\pi}{2}} x \log^2 \cos x dx$$

Note that

$$\begin{aligned}
\log\left(\frac{1+e^{2ix}}{2}\right) &= \log \cos x + ix \\
\Im(\log^3\left(\frac{1+e^{2ix}}{2}\right)) &= 3x \log^2 \cos x - x^3
\end{aligned}$$

We have

$$\begin{aligned}
\int_0^{\frac{\pi}{2}} x \log^2 \cos x dx &= \frac{1}{3} \int_0^{\frac{\pi}{2}} x^3 dx + \frac{1}{3} \Im \int_0^{\frac{\pi}{2}} \log^3\left(\frac{1+e^{2ix}}{2}\right) dx \\
&= \frac{\pi^4}{192} + \frac{1}{6} \Im \int_0^{\pi} \log^3\left(\frac{1+e^{ix}}{2}\right) dx \\
&= \frac{\pi^4}{192} - \frac{1}{6} \Re \int_{\Gamma} \frac{\log^3\left(\frac{1+z}{2}\right)}{z} dz
\end{aligned}$$

Γ represents the upper half of the unit circle, apply the Residue formula:

$$\begin{aligned}
\Re \int_{\Gamma} \frac{\log^3(\frac{1+z}{2})}{z} dz &= \int_0^1 \frac{1}{x} (\log^3(\frac{1-x}{2}) - \log^3(\frac{1+x}{2})) dx \\
&= 3 \log^2 2 \int_0^1 \frac{\log(1-x)}{x} dx - 3 \log 2 \int_0^1 \frac{\log^2(1-x)}{x} dx + \int_0^1 \frac{\log^3(1-x)}{x} dx \\
&\quad - 3 \log^2 2 \int_0^1 \frac{\log(1+x)}{x} dx + 3 \log 2 \int_0^1 \frac{\log^2(1+x)}{x} dx - \int_0^1 \frac{\log^3(1+x)}{x} dx \\
&= -\frac{2\pi^4}{15} - \pi^2 \log^2 2 + \frac{\log^4 2}{4} + 6\text{Li}_4\left(\frac{1}{2}\right)
\end{aligned}$$

Hence

$$\int_0^{\frac{\pi}{2}} x \log^2 \cos x dx = \frac{79\pi^4}{2880} + \frac{\pi^2 \log^2 2}{6} - \frac{\log^4 2}{24} - \text{Li}_4\left(\frac{1}{2}\right)$$

Since

$$\int_0^{\frac{\pi}{2}} x \log^2 \sin x dx = \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \log^2 \cos x dx - \int_0^{\frac{\pi}{2}} x \log^2 \cos x dx$$

Where

$$\begin{aligned}
\int_0^{\frac{\pi}{2}} \log^2 \cos x dx &= \int_0^{\frac{\pi}{2}} x^2 dx + \Re \int_0^{\frac{\pi}{2}} \log^2\left(\frac{1+e^{2ix}}{2}\right) dx \\
&= \frac{\pi^3}{24} + \frac{1}{2} \Im i \pi \log^2 2 \\
&= \frac{\pi^3}{24} + \frac{1}{2} \pi \log^2 2
\end{aligned}$$

We proved

$$\int_0^{\frac{\pi}{2}} x \log^2 \sin x dx = \text{Li}_4\left(\frac{1}{2}\right) - \frac{19\pi^4}{2880} + \frac{1}{24} \log^4 2 + \frac{1}{12} \pi^2 \log^2 2$$

A016

$$\int_0^1 \frac{\text{Li}_3(x)}{x\sqrt{1-x^2}} dx = \frac{1}{2} \text{Li}_4\left(\frac{1}{2}\right) + \frac{41\pi^4}{5760} + \frac{1}{48} \log^4 2 + \frac{1}{6} \pi^2 \log^2 2$$

Proof

$$\begin{aligned}
\int_0^1 \frac{\text{Li}_3(x)}{x\sqrt{1-x^2}} dx &= \frac{1}{2} \int_0^1 \int_0^1 \frac{1}{x\sqrt{1-x^2}} \frac{x \log^2 y}{1-xy} dy dx \\
&= \frac{1}{2} \int_0^1 \log^2 y \int_0^1 \frac{1}{(1-xy)\sqrt{1-x^2}} dx dy \\
&= \frac{1}{2} \int_0^1 \frac{\pi - \arccos y}{\sqrt{1-y^2}} \log^2 y dy \\
&= \frac{1}{2} \int_0^1 \frac{\pi - \frac{\pi}{2} + \arcsin x}{\sqrt{1-y^2}} \log^2 y dy \\
&= \frac{\pi}{4} \int_0^1 \frac{\log^2 y}{\sqrt{1-y^2}} dy + \frac{1}{2} \int_0^1 \frac{\arcsin x}{\sqrt{1-y^2}} \log^2 y dy \\
&= \frac{\pi^4}{96} + \frac{1}{8} \pi^2 \log^2 2 - \frac{1}{2} \int_0^1 \frac{\arcsin^2 y \log y}{y} dy \\
&= \frac{\pi^4}{96} + \frac{1}{8} \pi^2 \log^2 2 - \frac{1}{2} \int_0^{\frac{\pi}{2}} y^2 \log \sin y \cot y dy \\
&= \frac{\pi^4}{96} + \frac{1}{8} \pi^2 \log^2 2 + \frac{1}{2} \int_0^{\frac{\pi}{2}} y \log^2 \sin y dy \\
&= \frac{1}{2} \text{Li}_4\left(\frac{1}{2}\right) + \frac{41\pi^4}{5760} + \frac{1}{48} \log^4 2 + \frac{1}{6} \pi^2 \log^2 2
\end{aligned}$$

1.4 More Practices

B001

$$\int_0^1 \frac{\log(1-x) \log x}{1-x} dx$$

$$\begin{aligned} I &= \int_0^1 \log x d\left(-\frac{1}{2} \log^2(1-x)\right) \\ &= \frac{1}{2} \int_0^1 \frac{\log^2(1-x)}{x} dx \\ &= \frac{1}{2} \int_0^1 \frac{\log^2 x}{1-x} dx \\ &= \frac{1}{2} \int_0^\infty \frac{x^2}{e^x - 1} dx \\ &= \frac{1}{2} \Gamma(3) \zeta(3) \\ &= \zeta(3) \end{aligned}$$

B002

$$\int_0^1 \frac{\log x \log(1+x)}{1-x} dx$$

$$\begin{aligned} I &= \int_0^1 \frac{\log x \log(1-x^2)}{1-x} dx - \int_0^1 \frac{\log x \log(1-x)}{1-x} dx \\ &= \underbrace{\int_0^1 \frac{x \log x \log(1-x^2)}{1-x^2} dx}_{x^2 \rightarrow x} + \underbrace{\int_0^1 \frac{\log x \log(1-x)}{1-x^2} dx}_{x^2 \rightarrow x} - \zeta(3) \\ &= \frac{1}{4} \int_0^1 \frac{\log x \log(1-x)}{1-x} dx + \frac{1}{4} \int_0^1 \frac{x^{-\frac{1}{2}} \log x \log(1-x)}{1-x} dx - \zeta(3) \\ &= -\frac{3}{4} \zeta(3) + \frac{1}{4} \frac{\partial^2 B(p, q)}{\partial p \partial q} \Big|_{p=\frac{1}{2}, q \rightarrow 0^+} \\ &= \zeta(3) - \frac{\pi^2}{4} \log 2 \end{aligned}$$

B003

$$\sum_{n=1}^{\infty} \frac{H_n}{n^3}$$

(Euler Sum)

$$\begin{aligned}
S &= \sum_{n=1}^{\infty} \frac{1}{n^3} \int_0^1 \frac{1-t^n}{1-t} dt \\
&= \int_0^1 \left(\frac{1}{1-t} \sum_{n=1}^{\infty} \frac{1-t^n}{n^3} \right) dt \\
&= \int_0^1 \frac{\zeta(3) - \text{Li}_3(t)}{1-t} dt \\
&= -(\zeta(3) - \text{Li}_3(t)) \log(1-t) \Big|_0^1 - \int_0^1 \frac{\text{Li}_2(t) \log(1-t)}{t} dt \\
&= \int_0^1 \text{Li}_2(t) d(\text{Li}_2(t)) \\
&= \frac{\pi^4}{72}
\end{aligned}$$

B004

$$\sum_{n=1}^{\infty} \frac{H_n}{n^4 2^n}$$

Let

$$\mathcal{S} = \sum_{n=1}^{\infty} \frac{H_n}{n^4 2^n}$$

We first consider a slightly different yet related sum

$$\sum_{n=1}^{\infty} \frac{(-1)^n H_n}{n^4}$$

Note that

$$\begin{aligned}
\int_0^1 x^{n-1} \log^3 x dx &= \frac{\partial^3}{\partial n^3} \int_0^1 x^{n-1} dx = -\frac{6}{n^4} \\
\sum_{n=1}^{\infty} (-x)^{n-1} H_n &= \sum_{n=0}^{\infty} \sum_{k=0}^{\infty} (-x)^{n-k} \frac{(-x)^k}{k+1} = \frac{\log(1+x)}{x(1+x)}
\end{aligned}$$

We have

$$\begin{aligned}
\sum_{n=1}^{\infty} \frac{(-1)^n H_n}{n^4} &= \frac{1}{6} \int_0^1 \frac{\log^3 x \log(1+x)}{x(1+x)} dx \\
&= \frac{1}{6} \int_0^1 \frac{\log^3 x \log(1+x)}{x} dx - \frac{1}{6} \int_0^1 \frac{\log^3 x \log(1+x)}{1+x} dx
\end{aligned}$$

Therefore

$$\frac{1}{6} \int_0^1 \frac{\log^3 x \log(1+x)}{x} dx = \frac{1}{6} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \int_0^1 x^{n-1} \log^3 x dx = -\eta(5)$$

Apply a tricky transform:

$$\begin{aligned}
\frac{1}{6} \int_0^1 \frac{\log^3 x \log(1+x)}{1+x} dx &= \frac{1}{6} \int_1^2 \underbrace{\frac{\log^3(x-1) \log x}{x}}_{x-1 \rightarrow \frac{1-x}{x}} dx \\
&= \frac{1}{6} \int_{\frac{1}{2}}^1 \frac{\log x \log^3(\frac{x}{1-x})}{x} dx \\
&= \frac{1}{6} \int_{\frac{1}{2}}^1 \frac{\log x (\log x - \log(1-x))^3}{x} dx
\end{aligned}$$

Simplify after expanding:

$$\begin{aligned}
\sum_{n=1}^{\infty} \frac{(-1)^n H_n}{n^4} &= -\eta(5) + \underbrace{\int_{\frac{1}{2}}^1 \frac{\log x \log^3(1-x)}{6x} dx}_{\mathcal{I}_1} - \underbrace{\int_{\frac{1}{2}}^1 \frac{\log^2 x \log^2(1-x)}{2x} dx}_{\mathcal{I}_2} \\
&\quad + \underbrace{\int_{\frac{1}{2}}^1 \frac{\log^3 x \log(1-x)}{2x} dx}_{\mathcal{I}_3} - \underbrace{\int_{\frac{1}{2}}^1 \frac{\log^4 x}{6x} dx}_{\mathcal{I}_4}
\end{aligned}$$

Starting with the easiest integral,

$$\mathcal{I}_4 = \frac{1}{30} \log^5 2$$

For $\mathcal{I}_3$,

$$\begin{aligned}
\mathcal{I}_3 &= -\frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n} \int_{\frac{1}{2}}^1 x^{n-1} \log^3 x dx \\
&= -\frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n} \frac{\partial^3}{\partial n^3} \left(\frac{1}{n} - \frac{1}{n2^n} \right) \\
&= \sum_{n=1}^{\infty} \left(\frac{3}{n^5} - \frac{3}{n^5 2^n} - \frac{3 \log 2}{n^4 2^n} - \frac{3 \log^2 2}{n^3 2^{n+1}} - \frac{\log^3 2}{n^2 2^{n+1}} \right) \\
&= 3\zeta(5) - 3\text{Li}_5\left(\frac{1}{2}\right) - 3\text{Li}_4\left(\frac{1}{2}\right) \log 2 - \frac{3}{2} \log^2 2 \left(\frac{7}{8} \zeta(3) - \frac{\pi^2}{12} \log 2 + \frac{1}{6} \log^3 2 \right) \\
&\quad - \frac{1}{2} \log^3 2 \left(\frac{\pi^2}{12} - \frac{1}{2} \log^2 2 \right) \\
&= 3\zeta(5) - 3\text{Li}_5\left(\frac{1}{2}\right) - 3\text{Li}_4\left(\frac{1}{2}\right) \log 2 - \frac{21}{16} \zeta(3) \log^2 2 + \frac{\pi^2}{12} \log^3 2
\end{aligned}$$

For $\mathcal{I}_2$,

$$\begin{aligned}
\mathcal{I}_2 &= \frac{1}{6} \log^5 2 + \frac{1}{3} \int_{\frac{1}{2}}^1 \frac{\log^3 x \log(1-x)}{1-x} dx = \frac{1}{6} \log^5 2 - \frac{1}{3} \sum_{n=1}^{\infty} H_n \frac{\partial^3}{\partial n^3} \left(\frac{1}{n+1} - \frac{1}{(n+1)2^{n+1}} \right) \\
&= \frac{1}{6} \log^5 2 + \sum_{n=1}^{\infty} \frac{2H_n}{(n+1)^4} - \sum_{n=1}^{\infty} \frac{2H_n}{(n+1)^4 2^{n+1}} - \sum_{n=1}^{\infty} \frac{2 \log 2 H_n}{(n+1)^3 2^{n+1}} \\
&\quad - \sum_{n=1}^{\infty} \frac{\log^2 2 H_n}{(n+1)^2 2^{n+1}} - \sum_{n=1}^{\infty} \frac{\log^3 2 H_n}{3(n+1)2^{n+1}} \\
&= \frac{1}{6} \log^5 2 + 4\zeta(5) - \frac{\pi^2}{3} \zeta(3) - 2\mathcal{S} + 2\text{Li}_5 \left(\frac{1}{2} \right) - \frac{\pi^4}{360} \log 2 + \frac{1}{4} \zeta(3) \log^2 2 - \frac{1}{12} \log^5 2 \\
&\quad - \frac{1}{8} \zeta(3) \log^2 2 + \frac{1}{6} \log^5 2 - \frac{1}{6} \log^5 2 \\
&= -2\mathcal{S} + 2\text{Li}_5 \left(\frac{1}{2} \right) + 4\zeta(5) - \frac{\pi^4}{360} \log 2 + \frac{1}{8} \zeta(3) \log^2 2 - \frac{\pi^2}{3} \zeta(3) + \frac{1}{12} \log^5 2
\end{aligned}$$

For $\mathcal{I}_1$,

$$\begin{aligned}
\mathcal{I}_1 &= \frac{1}{6} \int_0^{\frac{1}{2}} \frac{\log^3 x \log(1-x)}{1-x} dx \\
&= -\frac{1}{6} \sum_{n=1}^{\infty} H_n \frac{\partial^3}{\partial n^3} \left(\frac{1}{(n+1)2^{n+1}} \right) \\
&= \sum_{n=1}^{\infty} \frac{H_n}{(n+1)^4 2^{n+1}} + \sum_{n=1}^{\infty} \frac{\log 2 H_n}{(n+1)^3 2^{n+1}} + \sum_{n=1}^{\infty} \frac{\log^2 2 H_n}{2(n+1)^2 2^{n+1}} + \sum_{n=1}^{\infty} \frac{\log^3 2 H_n}{6(n+1)2^{n+1}} \\
&= \mathcal{S} - \text{Li}_5 \left(\frac{1}{2} \right) + \frac{\pi^4}{720} \log 2 - \frac{1}{16} \zeta(3) \log^2 2 + \frac{1}{24} \log^5 2
\end{aligned}$$

Combining these four integrals gives

$$\begin{aligned}
\sum_{n=1}^{\infty} \frac{(-1)^n H_n}{n^4} &= 3\mathcal{S} - 6\text{Li}_5 \left(\frac{1}{2} \right) - \frac{31}{16} \zeta(5) - 3\text{Li}_4 \left(\frac{1}{2} \right) \log 2 + \frac{\pi^4}{240} \log 2 \\
&\quad - \frac{3}{2} \zeta(3) \log^2 2 + \frac{\pi^2}{3} \zeta(3) + \frac{\pi^2}{12} \log^3 2 - \frac{3}{40} \log^5 2
\end{aligned}$$

But consider $f(z) = \frac{\pi \csc(\pi z)(\gamma + \psi(-z))}{z^4}$.

At the positive integers,

$$\sum_{n=1}^{\infty} \text{Res}(f, n) = \sum_{n=1}^{\infty} \text{Res} \left(\frac{(-1)^n}{z^4(z-n)^2} + \frac{(-1)^n H_n}{z^4(z-n)} \right) = \sum_{n=1}^{\infty} \frac{(-1)^n H_n}{n^4} + \frac{15}{4} \zeta(5)$$

At $z = 0$,

$$\begin{aligned}
\text{Res}(f, 0) &= z^3 \left(\frac{1}{z} + \frac{\pi^2}{6} z + \frac{7\pi^4}{360} z^3 \right) \left(\frac{1}{z} - \frac{\pi^2}{6} z - \zeta(3) z^2 - \frac{\pi^4}{90} z^3 - \zeta(5) z^4 \right) \\
&= -\zeta(5) - \frac{\pi^2}{6} \zeta(3)
\end{aligned}$$

At the negative integers,

$$\sum_{n=1}^{\infty} \text{Res}(f, -n) = \sum_{n=1}^{\infty} \frac{(-1)^n H_n}{n^4} + \frac{15}{16} \zeta(5)$$

Since the sum of the residues is zero,

$$\sum_{n=1}^{\infty} \frac{(-1)^n H_n}{n^4} = -\frac{59}{32} \zeta(5) + \frac{\pi^2}{12} \zeta(3)$$

This implies that

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{H_n}{n^4 2^n} = & 2\text{Li}_5\left(\frac{1}{2}\right) + \frac{1}{32} \zeta(5) + \text{Li}_4\left(\frac{1}{2}\right) \log 2 - \frac{\pi^4}{720} \log 2 + \frac{1}{2} \zeta(3) \log^2 2 \\ & - \frac{\pi^2}{12} \zeta(3) - \frac{\pi^2}{36} \log^3 2 + \frac{1}{40} \log^5 2 \end{aligned}$$